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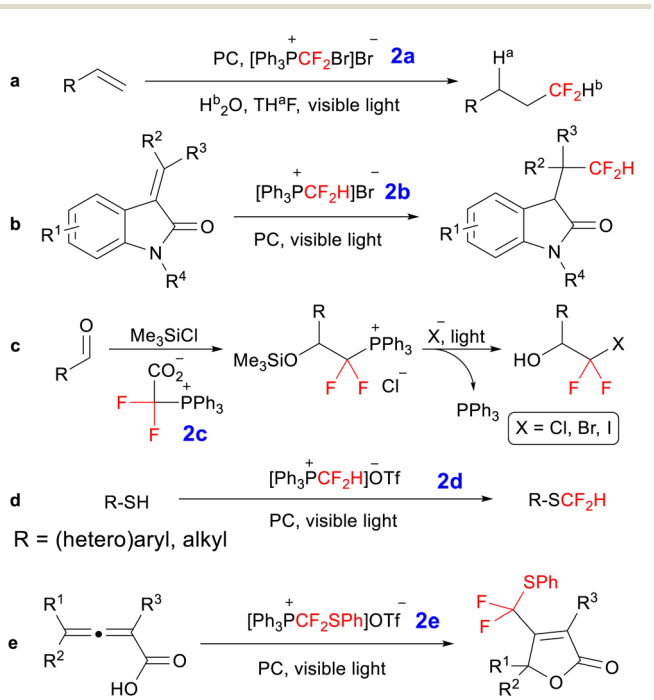
Photoredox/copper-catalyzed *gem*-difluoroalkylation-cyanation of 1,3-enynes†

Yachen Wang,[a] Shuai Liu[b] and Yangen Huang *[a]

A photoredox/copper-catalyzed 1,4-difunctionalization of 1,3-enynes with readily available difluoroalkylating reagents and TMSCN was developed. This reaction proceeded at mild conditions, affording the corresponding difluoroalkylated allenes in good yields with high functional-group tolerance and excellent regioselectivity.

The selective incorporation of *gem*-difluoroalkyl groups into drug-like molecules will have a wide impact on the physicochemical properties and biological activity of the parent molecules, so the development of efficient methods for the incorporation of *gem*-difluoroalkyl moieties into organic molecules has important value for medicinal chemistry and drug development.[1] In the past decades, visible-light photoredox catalysis has been widely used in the preparation of fluorinated compounds.[2] In 2016, Qing's group disclosed that fluorinated phosphonium salts **2a** could serve as the fluoroalkyl radical precursors[3] (Scheme 1a). Thenceforth fluorinated phosphonium salts have been widely applied in various radical fluoroalkylation reactions.[4] For example, Qing's group have developed a hydrodifluoromethylation of alkenes derived from oxindoles with (difluoromethyl)triphenylphosphonium bromide **2b** under photoredox catalysis[5] (Scheme 1b). Dilman's group have shown that halodifluoromethylated alcohols could be accessed from aldehydes and a readily available *gem*-difluoroalkylation reagent – phosphobetaine **2c** (Scheme 1c).[6] Meanwhile, Qing's group have disclosed visible light-induced radical *gem*-difluoromethylation of thiols with difluoromethyltriphenylphosphonium triflate **2d**[7] (Scheme 1d) and difluoro(phenylthio)methylation of 2,3-allenoic acids with {difluoro(phenylthio)-methyl}triphenylphosphonium triflate **2e** (Scheme 1e).[8]

Due to the unique structural characteristics,[9] allenes not only have important physiological and pharmacological activities, but also show very interesting reactivity and selectivity during chemical transformations.[10] Allenes are useful intermediates for the synthesis of natural products and drug molecules.[11] In the past few decades, the importance of allene chemistry has gradually been realized and attracted widespread attention, and methodologies for the synthesis and transformation of allenes have been well developed.[12] Thereinto, the 1,4-difunctionalization of 1,3-enynes provides a versatile and effective method for the synthesis of allenes.[13] The conventional approaches for 1,4-difunctionalization of 1,3-enynes focused on the allenyl anion pathway, in which



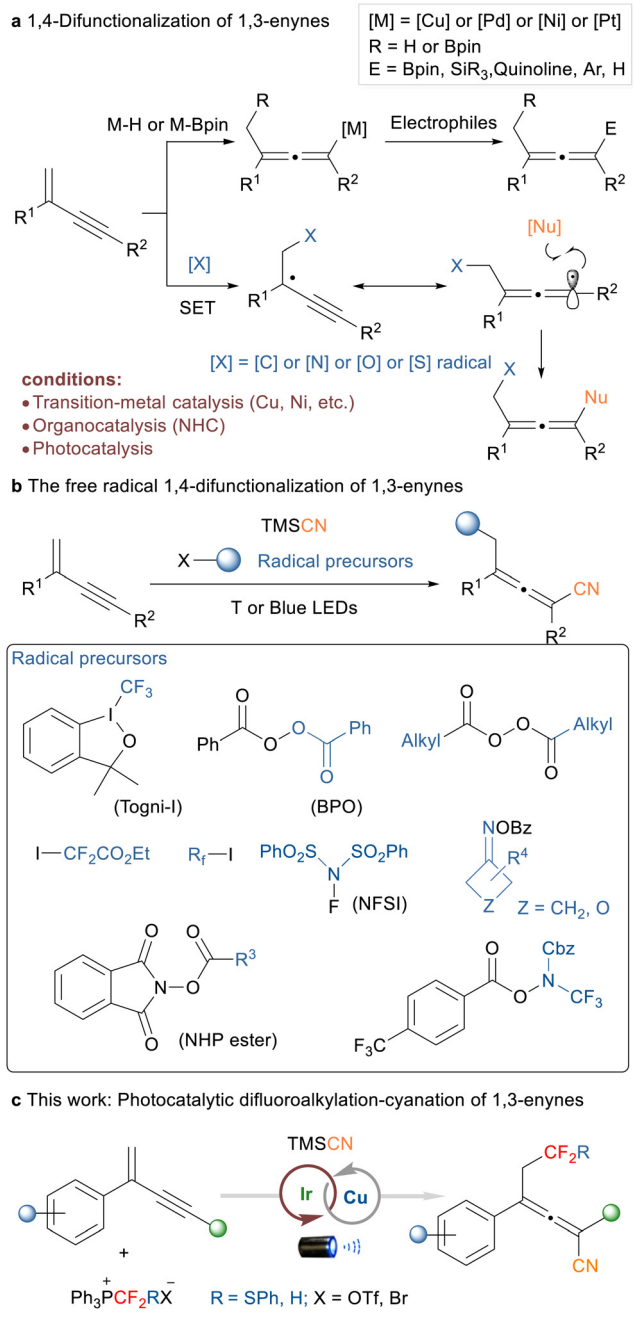
Scheme 1 Representive fluoroalkylphosphonium salts used in radical reactions.

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only electrophiles can be incorporated into the allene backbone in the second step of functionalization.[14] The 1,4-difunctionalization of 1,3-enynes *via* a radical pathway would complement conventional approaches for the synthesis of allenes (Scheme 2a). Three free radical pathways were mainly developed including low-valence metal such as Cu and Ni catalyzed single-electron reduction,[15] organic molecule catalysis (N-heterocyclic carbene)[16] and photocatalyzed single-electron oxidation or reduction.[17] In 2018, Liu's group reported Cu-catalyzed ligand-controlled radical 1,2- and 1,4-addition of 1,3-enynes provides access to a variety of CF_3-substituted allenyl nitriles with Togni-I as CF_3 source.[18] In addition to the use of Togni trifluoromethyl radicals, benzoyl peroxide (BPO),[14] fluoroalkylated iodides,[18] NFSI,[18,19] NHP ester,[20] cyclobutanone oxime esters[21] and *N*-Cbz-*N*-trifluoromethyl hydroxylamine reagents[17b,22] were applied as radical precursors to achieve the corresponding radical functionalization/cyanation reaction under different conditions (Scheme 2b).



Scheme 2 1,4-Difunctionalization of 1,3-enynes.

Inspired by these impressive achievements and recent advances, we envisioned that the *gem*-difluoroalkylallenes could be accessed *via* photoredox/copper-catalyzed *gem*-difluoroalkylation-cyanation of 1,3-enynes from TMSCN and difluoromethyl/difluoro(phenylthio)methyl phosphonium reagents (Scheme 2c). Considering the importance of difluoroalkyl groups and the potential derivatization of nitrile group,[23] the incorporation of both of these groups into allene molecules would be worthwhile.

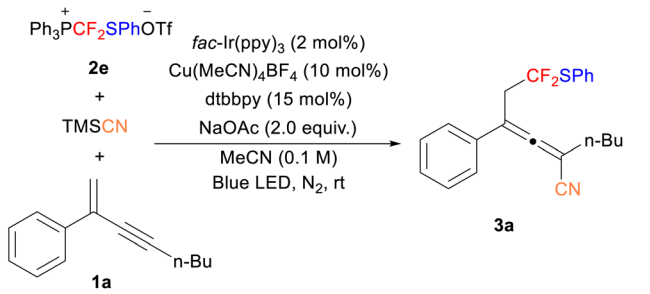
Our study commenced with oct-1-en-3-yn-2-ylbenzene (**1a**) as the model substrate, its reactivity with {difluoro(phenylthio)methyl}triphenylphosphonium triflate (**2e**) and TMSCN under photoredox/copper dual catalytic systems was investigated (Table 1). Gratifyingly, the desired 1,4-difunctionalized product **3a** was formed in 88% yield by using a catalytic system consisting of *fac*-Ir(ppy)$_3$ and Cu(MeCN)$_4$BF$_4$/dtbbpy with NaOAc as the base in MeCN under irradiation with blue LEDs (entry 1). None of the 1,2-difunctionalized product could be detected. In addition, switching MeCN to other solvents including PhCl, DCE, THF, DMF and EtOAc proved less effective (entries 2–6). Screening of different copper catalysts revealed that Cu(MeCN)$_4$BF$_4$ was optimal (entries 7–9). Subsequently, other bases including $NaHCO_3$, K_3PO_4, MeONa, CsF and NaOAc were evaluated (see the ESI†), and the best result was achieved when NaOAc was used as the base (entry 1). After a series of explorations (see the ESI†), it was found that the most appropriate ratio of reactants **1a**/**2e** is 1.5/1 (entries 13 and 14). Studies of reaction duration found that 6 h was more efficient. The effects of ligands on the reaction were also studied, and found that 15 mol% of dtbbpy was optimal (see the ESI†). Switching the photocatalyst from *fac*-Ir(ppy)$_3$ to Ir[dF(CF$_3$)ppy]$_2$(dtbbpy)PF$_6$, 9-mesityl-10-methylacridinium perchlorate, [Ir(ppy)$_2$(dtbbpy)]PF$_6$ or perylene led to lower yields, whereas the employment of Rhodamine 6G, Ru(bpy)$_3$Cl$_2$·6H$_2$O, 4CzIPN, thioxanthen-9-one, Ru(bpy)$_3$(PF$_6$)$_2$ and Solvent Red 43 as the photocatalyst could not afford any of the desired product (see the ESI†). The control experiments demonstrated that visible light, photocatalyst and Cu catalyst were indispensable for the reaction, while ligands and alkalis could promote the reaction (entries 17–22).

With the optimized reaction conditions in hand, the substrate scope of the reaction was investigated (Scheme 3). Various 1,3-enynes (**1a–y**) underwent this transformation smoothly, providing corresponding difluoro(phenylthiol)alkylated allenes (**3a–y**) in moderate to good yields. For 2-aryl-substituted 1,3-enynes, common functional groups on the benzene ring such as alkyl (**1b–f**), halogen (**1g–j**), alkoxy (**1k**), trifluoromethyl (**1l**), nitrile (**1m**), phenyl (**1n**) and benzyloxy

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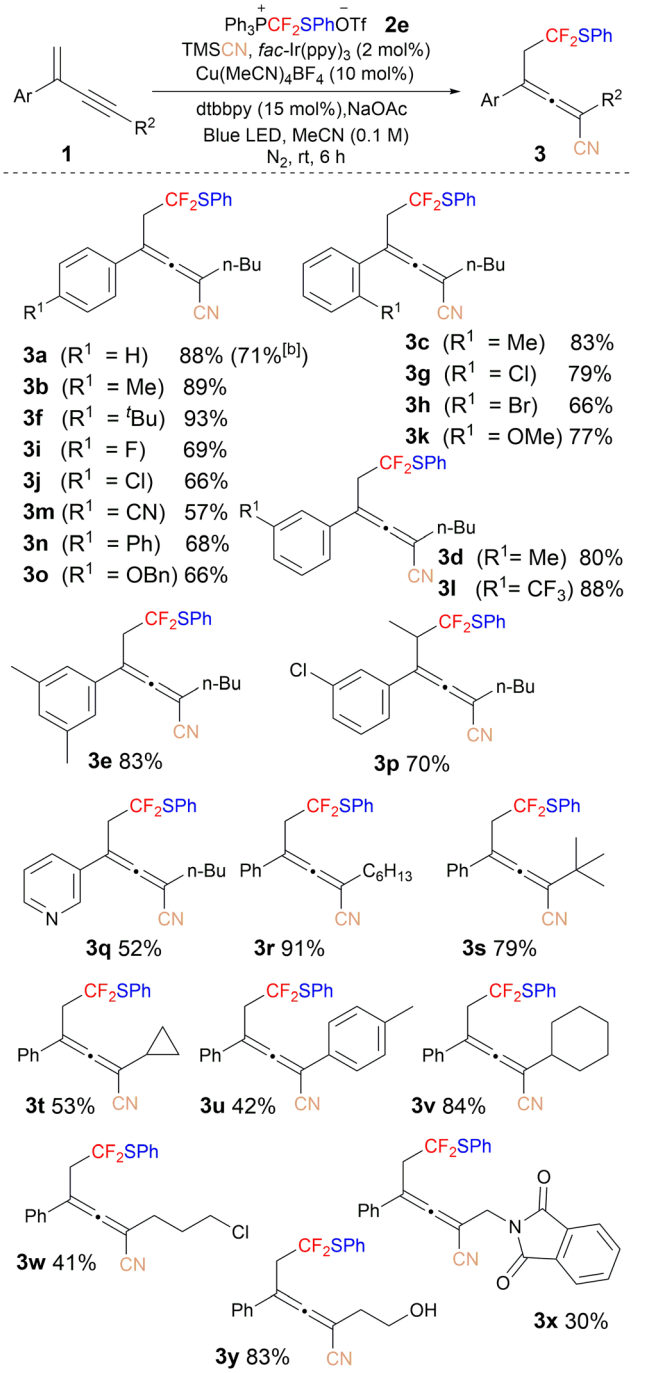
Table 1 Optimization of reaction conditions[a]



Entry	Variations from optimal conditions	Yield[b]
1	None	90, 88[c]
2	PhCl	<5
3	DCE	<5
4	THF	<5
5	DMF	<5
6	EtOAc	15
7	$Cu(MeCN)_4PF_6$	77
8	$Cu(OAc)_2$	39
9	CuCl	50
10	K_2CO_3	83
11	MeONa	32
12	CsF	53
13	1.0 equiv. **1a**, 1.5 equiv. **2e**	71
14	2.0 equiv. **1a**, 1.0 equiv. **2e**	86
15	4,4-D(Me)bpy	62
16	1,10-Phen	60
17	No *fac*-Ir(ppy)$_3$	0
18	No dtbbpy	63
19	No $Cu(MeCN)_4BF_4$	0
20	No NaOAc	36
21	In the dark	0
22	In air	<5

[a] Reaction conditions: **1a** (0.15 mmol), **2e** (0.1 mmol), TMSCN (0.2 mmol), photocatalyst (0.002 mmol), $Cu(MeCN)_4BF_4$ (0.01 mmol), dtbbpy (0.015 mmol), NaOAc (0.2 mmol), MeCN (1 mL), blue LED, 6 h, under N_2, rt. [b] Yields were determined by ^{19}F NMR spectroscopy with trifluorotoluene as an internal standard. [c] Isolated yield.

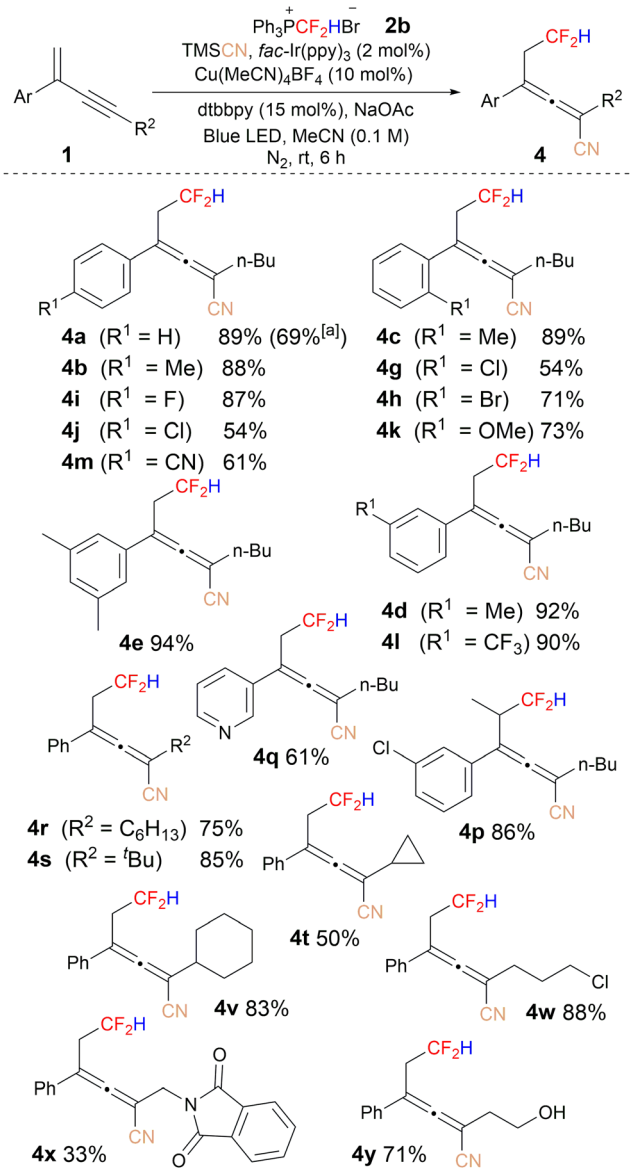
(**1o**) were well tolerated. 1,3-Enyne (**1e**) with two substituents on the benzene ring was efficiently converted to the corresponding product (**3e**). In addition, pyridyl-substituted 1,3-enyne (**1q**) was found to be a suitable substrate, affording the target product **3q** in moderate yield. 1,2,4-Trisubstituted 1,3-enyne (**1p**) is also compatible with this reaction. Next, the reactivity of various 1,3-enyne with different R^2 group were tested, such as hexyl (**1r**), *tert*-butyl (**1s**), cyclopropyl (**1t**), totyl (**1u**), cyclohexyl (**1v**), chloro (**1w**), amide (**1x**) and hydroxyalkyl (**1y**). These substrates are tolerated, providing various allenyl nitriles **3r–y** in moderate to good yields. The difluoroalkylation-cyanation of 1,3-enynes was further investigated by switching the quaternary phosphonium reagent from **2e** to (difluoromethyl)triphenylphosphonium bromide **2b** (Scheme 4). Fortunately, difluoromethylcyanoylated allenes **4a–y** were obtained in moderated to excellent yields (33–92%) under the optimal conditions. The reactions of **1a** could be easily scaled up to 3 mmol scale, affording **3a** and **4a** in yield of 71% and 69%, respectively.



Scheme 3 Substrate scope of difluoro(phenylthiol)methylation of 1,3-enyne. Reaction conditions: **1** (0.3 mmol), reagent **2e** (0.2 mmol), *fac*-Ir(ppy)$_3$ (0.004 mmol), $Cu(MeCN)_4BF_4$ (0.02 mmol), dtbbpy (0.03 mmol), NaOAc (0.4 mmol), TMSCN (0.4 mmol), MeCN (2.0 mL), blue LEDs, under N_2, rt, 6 h, isolated yields. [a]The reaction was performed in a 3 mmol scale.

In order to showcase the synthetic utility of the method, transformations of the allenyl nitriles were demonstrated (Scheme 5). **4a** could be readily converted into 6,6-difluoro-3-iodo-4-phenylhexa-2,4-dienenitrile **5** in 65% yield with 1.5 equiv. of *N*-iodosuccinimide (NIS) (Scheme 5a).[24]

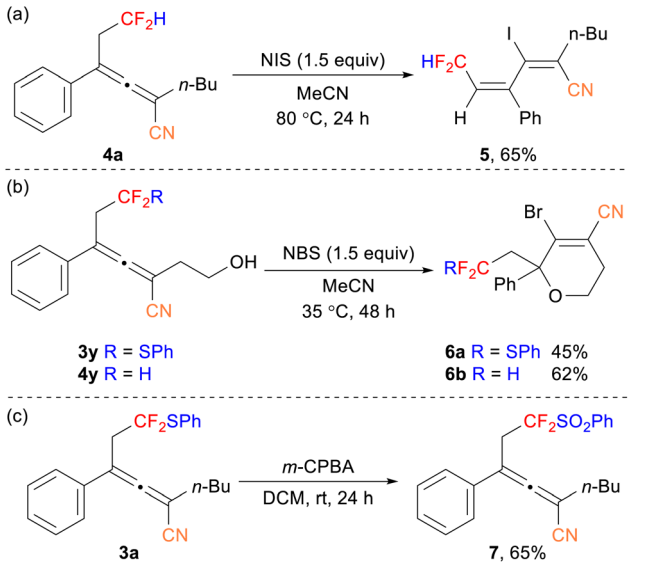
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Scheme 4 Substrate scope of difluoromethylation of 1,3-enyne. Reaction conditions: **1** (0.3 mmol), reagent **2b** (0.2 mmol), *fac*-Ir(ppy)$_3$ (0.004 mmol), Cu(MeCN)$_4$BF$_4$ (0.02 mmol), dtbbpy (0.03 mmol), NaOAc (0.4 mmol), TMSCN (0.4 mmol), MeCN (2.0 mL), blue LEDs, under N_2, rt, 6 h, isolated yields. [a]The reaction was performed in a 3 mmol scale.

Moreover, the bromination/cyclization of PhSCF$_2$-substituted allene **3y** and **4y** with *N*-bromosuccinimide (NBS) afforded the corresponding 3,6-dihydro-2*H*-pyran-4-carbonitrile **6a** and **6b** in yield of 45% and 62%, respectively. (Scheme 5b).[14,21,25] Furthermore, oxidation of **3a** with *m*-CPBA (3-chloroperoxybenzoic acid) gave PhSO$_2$CF$_2$-substituted allenyl nitrile **7** in 65% yield (Scheme 5c).[26]

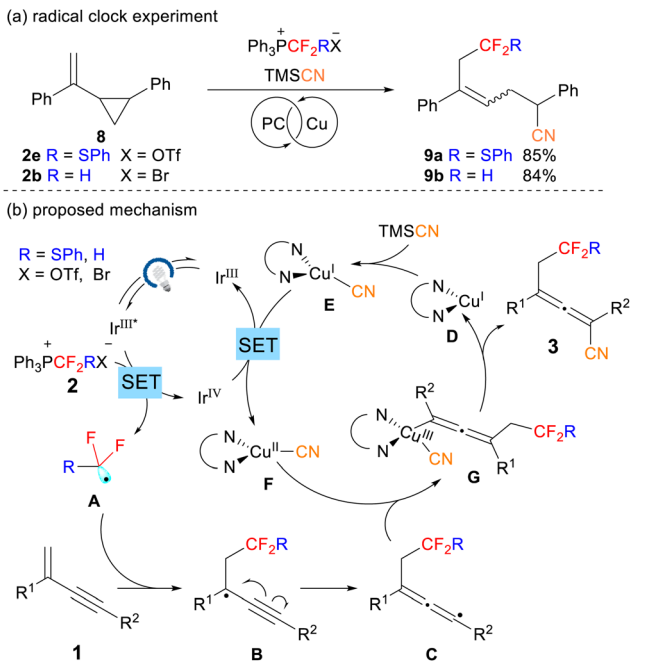
In order to gain mechanistic insight, several preliminary mechanistic experiments were performed. The absence of either the photoredox catalyst *fac*-Ir(ppy)$_3$ or visible light shut down the reactivity completely, thus suggesting a crucial role for both of these elements for the transformation (Table 1,



Scheme 5 Chemical transformation of difluoroalkyl allenes.

entries 17 and 21). The formation of desired product **3a** was completely inhibited under otherwise identical standard conditions after adding a radical scavenger (TEMPO, 2,2,6,6-tetramethylpiperidine-1-oxyl). This result demonstrated that a radical pathway was probably involved in this transformation.

Furthermore, a radical clock experiment with vinyl cyclopropane **8** was carried out to deliver the ring opening product **9a** and **9b** in yield of 85% and 84%, respectively (Scheme 6a). On the basis of this result and literature precedents,[2e,4f,14,15,17b,20–22,27] a plausible reaction mechanism is depicted in Scheme 6b.



Scheme 6 Mechanistic investigations.

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Firstly, photocatalyst *fac*-Ir(ppy)$_3$ is excited under visible-light irradiation. Then, single-electron transfer (SET) reduction of phosphonium reagent **2** with *fac*-Ir^{III}(ppy)$_3$* affords difluoroalkyl radical **A** and *fac*-Ir^{IV}(ppy)$_3$. Subsequently, difluoroalkyl radicals **A** were added to 1,3-enyne to form allenyl radical **B**, which was then isomerized to obtain radical intermediate **C**. Next, allenyl radical **C** were combined with [ligand/Cu^{II}(CN)] complex **F** to generate [Cu^{III}] complex **G**. Finally, reductive elimination of complex **G** releases the desired product **3** and regenerates the Cu^{I} catalyst.

Conclusions

In conclusion, we have developed a practical photoredox/copper-catalyzed difluoroalkylation-cyanation of 1,3-enynes with {difluoro(phenylthio)methyl}triphenylphosphonium triflate/(difluoromethyl)triphenylphosphonium bromide and TMSCN. This protocol provides a straightforward access to a series of previously unknown and potentially valuable difluoro(phenylthio)methylated and difluoromethylated allenes. Further investigations of the applications of fluoroalkylphosphonium salts in organic synthesis and medicinal chemistry are currently underway.

Author contributions

Yachen Wang: conceptualization, methodology, data curation, validation, writing – original draft. Shuai Liu: methodology, formal analysis. Yangen Huang: supervision, project administration, funding acquisition, writing – review & editing.

Conflicts of interest

There are no conflicts to declare.

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